

PCA 2--Lab-4 Transaction : COMMIT and ROLLBACK with/without SAVEPOINTS

1. BEGIN TRANSACTION;

//Perform database updates

```
UPDATE accounts SET balance = balance - 500 WHERE account_id = 1;
```

```
UPDATE accounts SET balance = balance + 500 WHERE account_id = 2;
```

// Permanently save changes

```
COMMIT;
```

2. Perform delete operation on any table of your choice. Use COMMIT as well as ROLL BACK to check the result.

3. Use of SAVEPOINT

```
-- 1. BEGIN TRANSACTION;
```

```
-- 2. Execute the first update
```

```
    UPDATE Customers SET State = 'California' WHERE ID = 1;
```

```
-- 3. Set a SAVEPOINT SAVEPOINT sp_after_first_update;
```

```
-- 4. Execute a second update UPDATE Customers SET State =  
'Nevada' WHERE ID = 2;
```

```
-- 5. Roll back only the second update (if needed) ROLLBACK TO  
SAVEPOINT sp_after_first_update;
```

```
-- 6. Commit the first update permanently COMMIT;
```